to be finished 4th on W

NOTE:

1. The line $x = a$ is a VA of the graph of $f(x)$ if a is the zero of the **simplified** denominator.

~~1~~2. A graph DOES NOT cross its vertical asymptote(s).

Factor, then simplify, ~~then~~

① **ex.** Find each vertical asymptote of the function.

1) $f(x) = \dfrac{x+2}{x^2-9} = \dfrac{x+2}{(x+3)(x-3)}$ VA $x=-3$, $x=3$

2) $g(x) = \dfrac{2}{x^2+4}$ simplified $= 0$ no real solutions, thus no VA.

3) $h(x) = \dfrac{x^2-2x}{x^2-3x+2} = \dfrac{x\cancel{(x-2)}}{(x-1)\cancel{(x-2)}} = \dfrac{x}{x-1}$

$D_h: x \neq 1, 2$

VA: $x=1$

Hole at $x=2$.

F12 4th CAR

Lecture 17: Section 2.6
Rational Functions

Holes and vertical asymptotes

Horizontal asymptotes

Graphing a rational function

Lecture 17: Section 2.6
Rational Functions

Holes and Vertical Asymptotes

The graph of a rational function $f(x) = \dfrac{p(x)}{q(x)}$ has a **hole** at $x = a$ if

1. both $p(x)$ and $q(x)$ have the common factor $(x - a)$, and
2. the **simplified** denominator DOES NOT have the factor $(x - a)$.

NOTE: To find the y-coordinate of the hole in the graph, plug $x = a$ into the **simplified** expression of $f(x)$.

ex. Find all vertical asymptotes and holes of the function:

Note!

D_f: $x \neq -1, 2$

Hole VA

1) $f(x) = \dfrac{x+1}{x^2 - x - 2} = \dfrac{\cancel{x+1}}{\cancel{(x+1)}(x-2)} = \dfrac{1}{x-2}$

$\boxed{\text{VA: } x = 2}$

Hole at $x = -1$

$\dfrac{1}{-1-2} = \dfrac{1}{-3}$

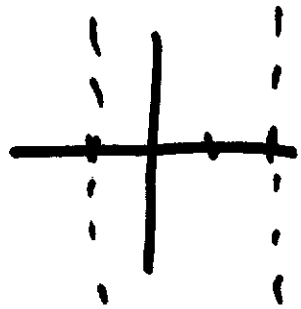
Hole $(-1, -\frac{1}{3})$

Note: D_g: $x \neq 0, \overset{VA}{-1}, \overset{VA}{2}, 3$

2) $g(x) = \dfrac{2x^{\cancel{6}\,3}\cancel{(x+1)}\cancel{(x-3)}}{\cancel{x}(x+1)^{\cancel{2}}(x-2)\cancel{(x-3)}} = \dfrac{2x^2}{(x+1)(x-2)}$

For VA

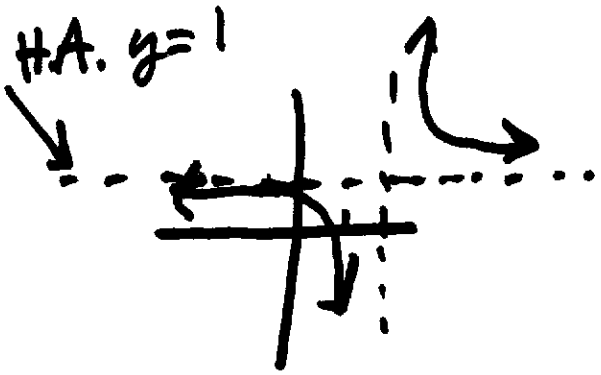
$\boxed{\text{VA: } x = -1 \quad x = 2}$



Holes at $x = 0$

$\dfrac{2x^2}{(x+1)(x-2)} \rightarrow \dfrac{0}{1 \cdot -2} = \dfrac{0}{-2} = 0$ $\boxed{(0,0) \text{ hole}}$

and $x = 3 \rightarrow \dfrac{2(3^2)}{(3+1)(3-2)} = \dfrac{18}{4} = \dfrac{9}{2}$ $\boxed{(3, \tfrac{9}{2}) \text{ hole}}$



$f(x) = \frac{1}{x-2} + 1$

Horizontal Asymptotes

<u>**Def.**</u> The line $y = L$ is a **horizontal asymptote (HA)** of the graph $y = f(x)$ if $f(x) \to L$ when $x \to \infty$ or when $x \to -\infty$

To Find Horizontal Asymptotes: * Look at lead terms of num. and den.

Let $f(x) = \dfrac{p(x)}{q(x)} = \dfrac{a_n x^n + \cdots + a_0}{b_m x^m + \cdots + b_0}$.

1. If $n < m$, $f(x)$ is a **proper** rational function. The graph of f has a horizontal asymptote $y = 0$

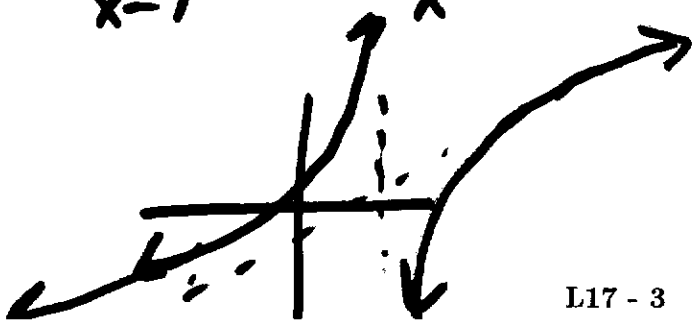
$f(x) = \dfrac{x + 100}{x^2 - 8x + 1}$ $\quad 1 < 2 \quad \approx \dfrac{x}{x^2} = \dfrac{1}{x}$

2. If $n = m$, the graph of $f(x)$ has a horizontal asymptote is $\boxed{\dfrac{a_n}{b_m}}$

$g(x) = \dfrac{3x^4 - 7x^3 + 2x}{6x^4 - 100x^2 + 5} \approx \dfrac{3x^4}{6x^4} = \dfrac{3}{6} = \dfrac{1}{2}$ $\to$ $y = \frac{1}{2}$ is the H.A.

3. If $n > m$, the graph of $f(x)$ has

$h(x) = \dfrac{x^2 - x - 2}{x - 1} \approx \dfrac{x^2}{x} = x$ $\quad$ $\boxed{\text{no H.A.}}$



(1) $\frac{\text{small}}{\text{large}}$ HA is $y=0$

(3) $\frac{\text{large}}{\text{small}}$ no HA.

(2) $\frac{\text{same}}{\text{same}}$ HA is $\frac{\text{coef}}{\text{coef}}$

ex. Find all horizontal asymptotes of the following:

1) $f(x) = \dfrac{x}{x^2+1} \approx \dfrac{x}{x^2} = \dfrac{1}{x}$

rule #1

$\boxed{\text{HA. is the line } y=0}$

2) $f(x) = \dfrac{x^2-4}{2x+1} \approx \dfrac{x^2}{2x} = \dfrac{x}{2}$ rule #3

thus $\boxed{\text{no H.A}}$

3) $f(x) = \dfrac{3x^2+9}{3-2x^2} \approx \dfrac{3\cancel{x^2}}{-2\cancel{x^2}} = \dfrac{3}{-2}$

$\boxed{\text{HA is the line } y = -\frac{3}{2}}$

rule #2

* **NOTE:** A graph can cross its horizontal asymptote!

To Graph a Rational Function:

1. Factor numerator and denominator.

2. Find the domain.

3. Reduce the fraction to lowest terms. From here on out use the reduced version.

4. Find each vertical asymptote.

5. Find the coordinates of any holes in the function.

6. Find the horizontal asymptote if there is one. Does the graph of the function cross its horizontal asymptote? If so, at what point?

7. Find any x-intercept and y-intercept.

8. Check for symmetry.

9. Use the zeros of both numerator and denominator of the reduced rational function to divide the number line into test intervals. Choose a point in each interval to determine where the graph lies above or below the x-axis.

10. Sketch the asymptotes and plot the points found above; graph the function.

(1) $f(x) = \dfrac{(2x-1)(x+4)}{(x+2)(x-1)}$

(2) Domain: $x \neq -2$
$x \neq 1$

(3) Already reduced.

(4) VA: $x = -2$, $x = 1$

<u>ex.</u> Sketch the graph of $f(x) = \dfrac{2x^2+7x-4}{x^2+x-2}$.

(5) No holes, since no factors cancelled.

(6) $f(x) \approx \dfrac{2x^2}{x^2} = 2$ HA is the line $y = 2$

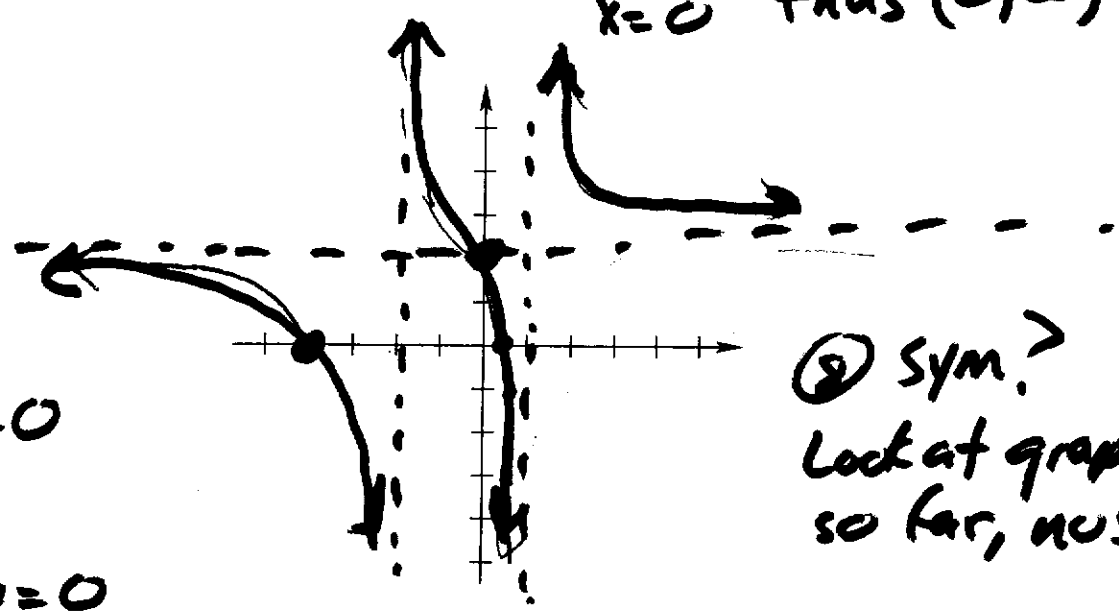
Do we cross the H.A.? Let $y = 2$

$\dfrac{2x^2+7x-4}{x^2+x-2} = 2$, $\cancel{2x^2}+7x-4 = \cancel{2x^2}+2x-4$

$5x = 0$
$x = 0$ thus $(0,2)$

(7)

x	y
1	0
2	
-4	0
0	2



$\dfrac{2x^2+7x-4}{x^2+x-2} = 0$

$2x^2+7x-4 = 0$

$(2x-1)(x+4) = 0$

Let $x = 0$

$f(0) = 2$

(8) Sym.?
Look at graph so far, no sym.

-4
-3
-2
-1
+
-
+
-
+
x = -5
x = -3
x = 0
x = 2
x = 5

① $f(x) = \dfrac{(x+1)(x-1)}{(x+2)(x-2)}$

③ reduce, doesn't reduce

④ V.A.: $x = -2$
$x = 2$

② D: $x \neq -2, 2$

ex. Sketch the graph of $f(x) = \dfrac{x^2-1}{x^2-4}$.

⑤ No holes.

⑥ HA. $\dfrac{x^2}{x^2} = 1$ HA is line $y = 1$.

cross HA? $\dfrac{x^2-1}{x^2-4} = 1$, $\cancel{x^2}-1 = \cancel{x^2}-4$

$-1 = -4$ thus no sol. so it doesn't cross HA.

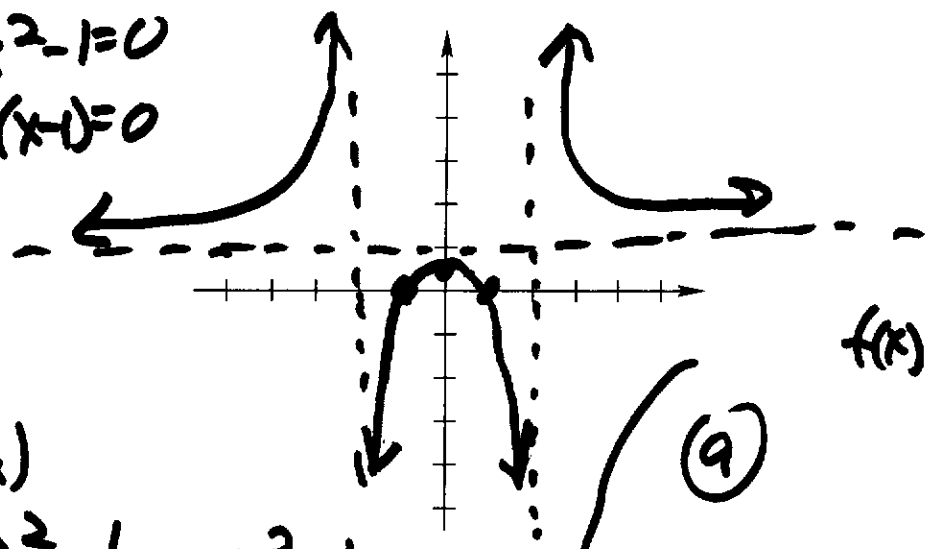
⑦

x	y
1	0
-1	0
0	$\frac{1}{4}$

$\dfrac{x^2-1}{x^2-4} = 0$

$x^2 - 1 = 0$

$(x+1)(x-1) = 0$



⑧ sym

even?

$f(-x) \overset{?}{=} f(x)$

$f(-x) = \dfrac{(-x)^2-1}{(-x)^2-4} = \dfrac{x^2-1}{x^2-4} = f(x)$

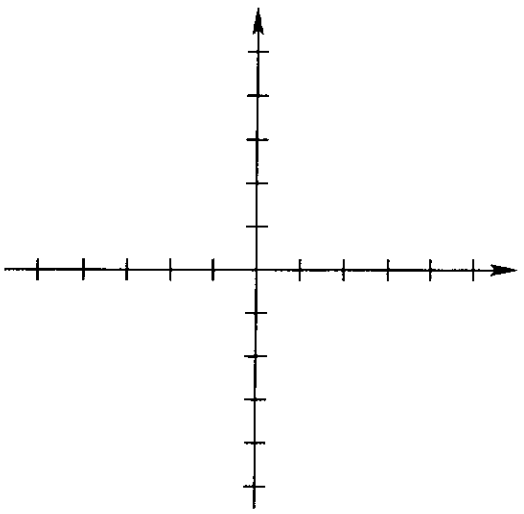
yes sym wrt y-axis

⑨ $f(x) = \dfrac{(x+1)(x-1)}{(x+2)(x-2)}$

− − −	− − −	+ − −	+ + +	+ + +
− − −	+ − −	+ − −	+ + −	+ + +
⊕	⊖	⊕	⊖	⊕

$-2 \quad -1 \quad 1 \quad 2$

ex. Sketch the graph of $f(x) = \dfrac{2x^2 - 4x}{x^2 + x - 6}$.



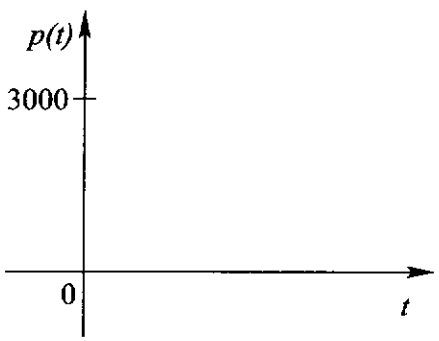
Applications

ex. The rabbit population on Mr. Jenkins' farm follows the model

$$p(t) = \frac{3000t}{t+1}$$

where $t \geq 0$ is the time (in months) since the beginning of the year.

1) Draw a graph of the rabbit population.



2) What eventually happens to the rabbit population?